SUMIT SINGH

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OBJECTIVE

AI/ML and Software Development professional with 4+ years of industry experience spanning applied ML, NLP, and Conversational AI in production systems. Currently pursuing an MSc in Applied Computing (AI) at the University of Toronto and seeking research-oriented opportunities in deep learning, generative AI, and foundation models.

EDUCATION

Master of Science in Applied Computing, *AI Concentration* Sep 2026 - Dec 2027
University of Toronto
Relevant Coursework: Neural Networks and Deep Learning; Imitation Learning for Robotics

Bachelor of Technology, Computer Science Engineering *(Specialization in AI & ML)* Jul 2018 - May 2022
SRM Institute of Science and Technology CGPA - 9.71/10.00

PROFESSIONAL EXPERIENCE

Goldman Sachs — Associate (SDE-2) Jul 2022 - Jul 2026

• **Conversational AI-based IVR:** Developed and deployed an IVR using **Amazon Lex** and **AWS Lambda (Go)**, with Java-based microservices to replace DTMF navigation with natural-language interaction across **60+ customer intents**; upgraded the Lex integration from ASR 1.0 to **ASR 2.0** to improve transcription and intent classification.
• **BERT-based Intent Classification:** Developed an in-house BERT-based intent classifier using production utterances mapped to business-specific intents, and applied **parameter-efficient fine-tuning** to improve classification.
• Increased IVR self-service containment **from 10% to 75% across approximately 600K calls per month**, reducing transfers to human agents and contributing to **$5M+ in quarterly savings**.
• **Automated Bot Regression Testing:** Engineered a serverless regression testing solution using **AWS Lambda and Step Functions**, processing **3,000+ test utterances weekly** and generating reports on model and bot issues.

Sabre — Associate Intern Jan 2022 - Jul 2022

• **Automated Python Data Scraping:** Developed Python-based web scraping scripts to collect airline data from public websites, and prepared the data in a structured format for downstream analysis.
• **Airline Data Analytics:** Prepared and formatted airline data for an internal analytics and optimization tool, supporting analysis of flight scheduling, fleet allocation, and dynamic pricing for client airlines.

HighRadius — Software Engineering Intern Aug 2021 - Jan 2022

• Developed **50+ JUnit test cases**, resolving edge cases and raising coverage from 0% to **83%**, ensuring reliability.
• Refactored Java codebase, improving maintainability, scalability and reusability across components.

PERSONAL PROJECTS

• **PhysicsTeacher-CurrentElectricity-GPT** 🔗: Built a physics tutoring system for Grade 11 Current Electricity using **RAG and ChromaDB**, with textbook retrieval and **embedding-based conversation memory**. Used **OpenAI** generation and **NVIDIA NeMo Guardrails** for grounded, domain-constrained responses.

• **ML-based Invoice Management Application** 🔗: Preprocessed **45K+** invoices, engineered **6 payment-trend features**, compared models, and tuned XGBoost via **5-fold cross-validation**, yielding **2.43-day MAE**, **0.61 R^2**.

KEY SKILLS

Languages: Python, Go, C++, Java, SQL, JavaScript
ML/AI: Fine-tuning, NLP, RAG, Prompt Engineering, NumPy, Pandas, Scikit-learn, XGBoost, TensorFlow
Tools: PyTorch, Hugging Face Transformers, ChromaDB, NVIDIA NeMo Guardrails, Amazon Lex, AWS Lambda

EXTRA-CURRICULAR ACTIVITIES

Authored **8 project tutorial blogs** on **CodeSpeedy.com**, generating **5,500+** cumulative views. 🔗